FLOOD GUARD 360 INTEGRATED RESEARCH PROJECT

COMMON INTEGRATED SOLUTION REPORT

Team member	Student ID	Individual component
SOMARATHNE B D	IT22091734	Flood Map Creation
BULUMULLA D M D B	IT22278258	Real Time Drain Water Level Monitoring, Rainfall Monitoring, and Multi Point Blockage Detection for Flood Risk Management
JAYANATH H P K N	IT22170002	To design and implement a post flood disease management module that enables early detection of flood related health risks using symptom data and facial scan analysis.
RASHMI P M U	IT22130198	To develop a post flood rescue and ration management system.

Report submitted in partial fulfilment of the requirements for the

BSc (Hons) Degree in Information Technology

Specialization in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

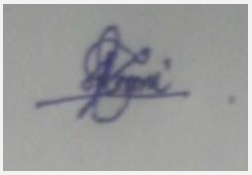
Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

Academic integrity and AI disclosure. We have disclosed all use of generative AI and other automated tools in accordance with the module policy. We remain responsible for the accuracy, originality, citations, code, data, analyses, and conclusions in this report.

Student name	Student ID	Signature	Date
SOMARATHNE B D	IT22091734		10/11/2026
BULUMULLA D M D B	IT22278258		10/11/2026
JAYANATH H P K N	IT22170002		10/11/2026
RASHMI P M U	IT22130198		10/11/2026

Supervisor certification. The above candidates carried out this research project under my supervision.

Role and name	Signature	Date
Supervisor: Mr. Deemantha Siriwardana		10/11/2026
Co-supervisor: Ms. Ayesha Wijesooriya		

ABSTRACT

Flooding can cause damage to homes, roads, public services, livelihoods, and community health. The effects may continue after floodwater recedes because blocked drainage, unsafe water, disease risks and urgent rescue or relief needs may remain. Flood Guard 360: 24/7, Every Hour is a proposed integrated system that connects four related functions: flood-risk map creation, real-time drain monitoring, post-flood disease-risk screening, and rescue and ration-distribution prioritization. The flood-risk mapping component combines geographic and environmental information to display locations that may be more exposed to flooding. The drain-management component uses ESP32-based devices, ultrasonic water-level sensors, and a rainfall sensor to collect readings. A separate multi-point monitoring unit compares readings from three positions to identify patterns that may indicate blockage. Sensor readings are sent to backend API endpoints for storage and display. The disease-screening component is intended to support preliminary risk assessment using user-reported symptoms and other approved information; it is not a medical diagnosis. The rescue and ration-management component is intended to classify affected locations by priority and help responders organize limited resources. The common design uses shared location identifiers, timestamps, data validation, and role-appropriate dashboards to connect information across components. Development and evaluation should combine component tests with end-to-end checks of data transfer, system usability, reliability, privacy, and the usefulness of outputs to intended users. As implementation and validation status differ between components, only measured results supported by test records should be reported as achieved outcomes. The integrated system aims to provide a clearer flow of information from flood risk awareness to monitoring and post-flood response, while requiring further field testing and stakeholder validation before operational deployment.

Keywords: flood risk management, Internet of Things, GIS, disease-risk screening, disaster response

ACKNOWLEDGEMENTS

We would like to express our sincere gratitude to my supervisor, Mr. Deemantha Siriwardhana, for his guidance and academic support during this final research and development work. We also acknowledge the support of the Flood Guard 360 research team and the academic staff of the Sri Lanka Institute of Information Technology. Their collaboration and feedback contributed to the broader integrated flood management project that was developed. We remain responsible for verifying the accuracy of the content, implementation details, data, results, and references presented in this thesis.

TABLE OF CONTENTS

1 INTRODUCTION AND PROJECT CONTEXT 1

1.1 Background and context 1

1.2 Stakeholders and user needs 2

1.3 Alignment with the Sustainable Development Goals 3

1.4 Purpose and scope of the common report 3

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS 4

2.1 Overall problem definition 4

2.2 Root causes and overall research gap 4

2.3 Subproblem and component mapping 5

2.4 Overall aim and group objectives 6

2.5 Scope assumptions and constraints 7

3 OVERALL SOLUTION STRUCTURE 9

3.1 Integrated solution overview 9

3.2 High level architecture 9

3.3 Component 1 summary- Flood Risk Map Creation 10

3.4 Component 2 summary - Real-Time Drain Monitoring and Blockage Detection 10

3.5 Component 3 summary - Post-Flood Disease Detection 10

3.6 Component 4 summary - Post-Flood Rescue and Ration Distribution Management 11

3.7 End to end user and data flow 11

4 COMPONENT INTEGRATION AND COHESIVE OPERATION 13

4.1 Integration strategy 13

4.2 Interfaces data contracts and shared models 14

4.3 Shared nonfunctional requirements 16

4.4 Integrated workflow and exception handling 17
4.5 Team contribution and dependency matrix 19
5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION 20
5.1 Common technology stack and infrastructure 20
5.2 Development and integration process 21
5.3 Deployment and operational configuration 22
6 INTEGRATED TESTING EVALUATION AND VALIDATION 23
6.1 Evaluation framework and success criteria 23
6.2 Integration and end to end testing 23
6.3 Overall results 26
6.4 Stakeholder and user validation 29
6.4.1 Feedback collection method 29
6.4.2 Evaluation results 30
6.4.3 Improvements identified through feedback 30
6.4.4 Limitations of the evaluation 31
6.5 Achievement of common objectives 32
6.6 Integrated limitations and threats 34
7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION 38
7.1 Value proposition and beneficiaries 38
7.2 Market alternatives and differentiation 39
7.3 Business model and feasibility 40
7.4 Adoption intellectual property and compliance 41
7.5 Sustainable Development Goal contribution 42
8 CONCLUSION AND FUTURE DIRECTIONS 44
8.1 Overall conclusion 44

8.2 Principal integrated contribution 44
8.3 Future directions 45
REFERENCES 48
APPENDICES 51
Appendix A Integrated interface specification 51
Appendix B Team contribution declaration 58

LIST OF FIGURES

Figure 1 : System Architecture 9
Figure 2 : Data Flow 12

LIST OF TABLES

Table 1 : Abbreviations x
Table 2 : Stakeholders 2
Table 3 : Component Mapping 5
Table 4 : Data Contracts 15
Table 5 : Nonfunctional requirements 16
Table 6 : Team Contribution 19
Table 7 : Common Technology 20
Table 8 : Integration test case 24
Table 9 : Results 26
Table 10 : Achievements 32
Table 11 : Main areas of differences 40
Table 12 : Integrated technology stack 51
Table 13 : Common interface requirements 54
Table 14 : Proposed integration verification tests 56
Table 15 : Individual Contribution 58

LIST OF ABBREVIATIONS

Table 1 : Abbreviations

Abbreviation	**Meaning**
API	Application Programming Interface
GIS	Geographic Information System
GPS	Global Positioning System
IOT	Internet of Things
ML	Machine Learning
SDG	Sustainable Development Goal
UI	User Interface
UX	User Experience
HTTP	Hypertext Transfer Protocol
ESP32	Espressif Systems 32-bit microcontroller family
RF	Random Forest

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Flooding is a complex public safety problem. Heavy rainfall can exceed the capacity of drains, canals, and other water-management infrastructure. Waste, sediment, and physical obstructions may reduce drainage flow and increase local water levels. Flood impacts are not limited to the period when water is rising. After an event, affected communities may face unsafe water, possible disease exposure, damaged access routes, and urgent needs for rescue, food, water and other supplies.

Different organizations may hold different parts of the information needed to respond to. Geographic information can show where risk may be higher, while sensors can provide current local observations. Health related reports can help authorized personnel recognize possible post flood risks, and rescue or relief teams need a way to compare the urgency of requests. If these activities are handled separately, information may be delayed, duplicated, or difficult to compare.

Flood Guard 360: 24/7, Every Hour is designed as an integrated research project to connect these four needs. It is not intended to replace official flood warnings, clinical assessment, or emergency service judgement. Instead, it aims to provide decision support information that can be reviewed by residents, relevant authorities, and response teams.

1.2 Stakeholders and user needs

Table 2 : Stakeholders

Stakeholder	Main information need	Expected system support
Residents and community members	Understand possible flood exposure and receive timely, understandable information	Risk map views and, where implemented, clear monitoring alerts and safety information
Local authorities and drainage teams	Identify locations with rising water levels or possible blockage	Time stamped sensor readings, status indicators and history
Disaster management officers	Compare risk information across locations	Map-based overview and prioritized location summaries
Public health staff	Review possible post-flood health concerns without treating automated outputs as diagnoses	Structured symptom reports and preliminary risk flags for authorized review
Rescue and relief coordinators	Identify urgent cases and organize available resources	Priority lists, location information and allocation records
System administrators	Maintain users, data quality and service availability	Access control, logs, monitoring and recovery procedures

1.3 Alignment with the Sustainable Development Goals

The project has a potential relationship with Sustainable Development Goal 11 (Sustainable Cities and Communities), because it focuses on disaster risk awareness, urban drainage, and support for safer communities. It also relates to SDG 13 (Climate Action) through its intended contribution to climate-related hazard preparedness and resilience. The health screening element has a potential relationship with SDG 3 (Good Health and Well-being), while the rescue and ration distribution element may support the wider aim of reducing hardship during emergencies.

1.4 Purpose and scope of the common report

The purpose of this common report is to present the overall design and development of Flood Guard 360: 24/7, Every Hour, an integrated system developed to support flood risk management and post flood response. The report explains the common problem addressed by the project, the proposed solution, and how the four components work together to support flood monitoring, risk identification, health-risk screening, and rescue and ration distribution management.

The scope of this report includes the overall system architecture, the main functions of each component, data sharing and integration, implementation approaches, testing and evaluation, and the potential benefits and limitations of the complete system. The four components are Flood Risk Map Creation, Real Time Drain Water Level and Rainfall Monitoring with Multi Point Blockage Detection, Post Flood Disease Detection, and Post Flood Rescue and Ration Distribution Management.

This report focuses on the relationship between the four components and their contribution to the overall system. Detailed technical designs, implementation methods, experiments, and component specific results are discussed in the individual theses of the respective team members. The common report brings these contributions together to explain the integrated solution, evaluate its overall progress, and identify areas for future improvement.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

Communities and response organizations may not have one connected view of flood related risk, current drainage conditions, possible post flood health concerns, and the priority of rescue or relief needs. Information can be distributed across separate processes, and some decisions may depend on delayed reports or manual coordination. The research problem is how to design and evaluate an integrated information system that brings these different types of information together while preserving data quality, safety, privacy, and human oversight.

2.2 Root causes and overall research gap

The problem has several related causes. First, flood risk varies by location and depends on geography, rainfall, drainage capacity, and historical events. Second, local drain conditions can change quickly, and a single water level reading may not explain why the level is rising. Third, health concerns can emerge after floodwater affects homes and public spaces, but automated screening must be used carefully because symptoms have many possible causes. Fourth, rescue and ration distribution require coordination across locations and limited resources.

A system that only displays a map does not necessarily provide live drain observations. A sensor only system may identify a local condition but may not explain wider geographic exposure. Health reporting and relief prioritization tools address later stages of the event, but they need reliable location information, access controls and clear handover between responsible users. The integrated research gap addressed by this project is the design of a shared workflow connecting flood risk awareness, local monitoring, preliminary health risk reporting, and post flood response support.

2.3 Subproblem and component mapping

Table 3 : Component Mapping

Component	Subproblem	Inputs	Intended output	Dependency
1. Flood Risk Map Creation	Users need a geographic view of locations that may be exposed to flooding	GIS layers, elevation, rainfall, historical flood records and other approved environmental data	Map with risk categories and supporting context	Provides location and risk context to the other components
2. Drain Management	Drain water-level changes, rainfall and possible obstructions may not be visible in real time	Ultrasonic sensor readings, rainfall readings, sensor identity, timestamp and location/package ID	Live readings, trends and possible blockage flags	Provides local observations that can complement map-based risk information
3. Post-Flood Disease Detection	Post-flood health concerns may need structured reporting and review	User-reported symptoms and approved contextual information	Preliminary risk category or referral prompt for authorized review	Uses location and event context where appropriate; health data must be access-controlled
4. Rescue and Ration	Response teams need to compare	Affected-location reports, risk context,	Priority classifications, task lists and	Uses outputs from other components

Distribution Management	needs and coordinate limited resources	requests and available resource information	distribution records	when available, but must handle missing data

2.4 Overall aim and group objectives

Overall aim: To design and evaluate an integrated flood management decision support system that connects geographic flood risk information, real time drain monitoring, preliminary post flood health risk screening and rescue or ration distribution prioritization.

- Develop a geographic representation of flood risk information using documented data sources and a stated mapping method.
- Collect and transmit drain water level and rainfall readings and evaluate sensor and API behavior under defined test conditions.
- Implement and assess a multi-point comparison method that flags readings consistent with a possible blockage, while recording false positives and sensor failures.
- Design a privacy aware workflow for structured post flood symptom reporting and preliminary risk screening, with human review and no claim of medical diagnosis.
- Develop a method for classifying affected locations by rescue or ration priority and document the model inputs, evaluation method and limitations.
- Define shared location identifiers, timestamps, data formats and interface responsibilities so that components can exchange information consistently.
- Evaluate end to end operation, usability, security, reliability and limitations using repeatable tests and recorded evidence.

2.5 Scope assumptions and constraints

The development of Flood Guard 360: 24/7, Every Hour is based on several assumptions and constraints that define the scope of the project.

Assumptions

- The system assumes that IoT sensors can collect water-level and rainfall readings and send them to the backend through a network connection.
- The flood risk map component assumes that relevant geographical, rainfall, and historical flood data are available to identify areas that may be at risk.
- The drain monitoring component assumes that ultrasonic sensors installed at different points can help identify unusual water-level differences that may indicate a possible blockage.
- The post-flood disease detection component assumes that users can provide relevant symptom information and that the system can use this information to identify possible health risks. The results are intended to support awareness and response, not replace professional medical diagnosis.
- The rescue and ration distribution component assumes that information from the other components can be used to help identify affected areas and support rescue and resource allocation decisions.
- The system assumes that relevant users, such as residents and responsible authorities, can access the dashboard and alerts when the required network and services are available.

Constraints

- The accuracy of flood-risk predictions depends on the quality, availability, and reliability of the collected data.
- IoT sensor readings may be affected by sensor placement, environmental conditions, calibration errors, and hardware limitations.
- Internet or power failures may interrupt real time data transmission, dashboard updates, and alert delivery.

- The prototype is developed within the available time, budget, hardware, and software resources. Therefore, testing may be limited to the prototype setup and selected test scenarios.
- The three-point blockage detection method identifies possible blockages based on water level differences. Further physical inspection may be required to confirm a blockage.
- Disease-risk screening depends on the information provided and the performance of the implemented analysis methods. It cannot confirm a disease.
- Rescue priority classifications are intended to support decision making and may not represent every situation on the ground. Final rescue and ration distribution decisions should consider verified field information and the judgement of responsible authorities.
- The overall system's performance depends on the successful integration of the four components. Differences in data formats, communication failures, or unavailable services may affect how information is shared.

3 OVERALL SOLUTION STRUCTURE

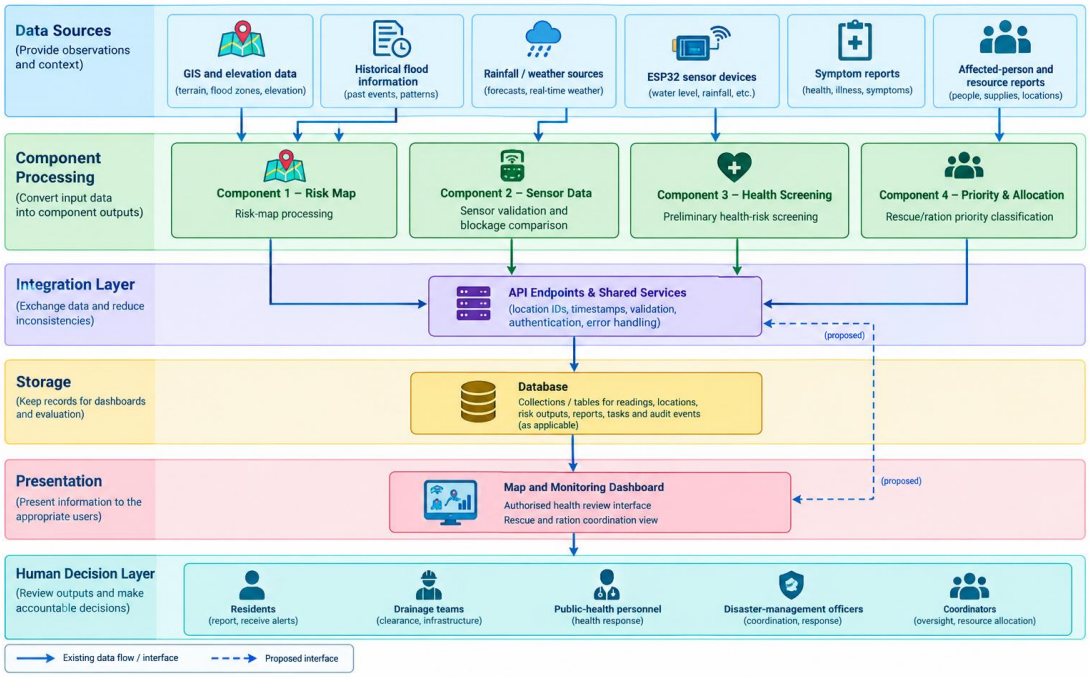
3.1 Integrated solution overview

Flood Guard 360 is organized around four components connected through shared location and event information. The geographic component gives a spatial overview of risk. The drain management component contributes to current sensor observations and possible blockage indicators. The health component is intended to collect and review possible post flood health concerns. The rescue and ration component is intended to help authorized responders compare reported needs and organize response actions.

The integrated value comes from the relationship between the components, rather than from treating them as four unrelated applications. For example, a location on the risk map may also have recent drain readings; after a flood, authorized teams may review health reports and response requests for that location. This example describes the intended workflow.

3.2 High level architecture

Figure 1 : System Architecture



3.3 Component 1 summary- Flood Risk Map Creation

The Flood Risk Map Creation component identifies areas that may be at risk of flooding by using geographical information, rainfall data, historical flood records, and other relevant environmental information. The component displays flood-risk areas on a map to help users understand which locations may be more vulnerable during heavy rainfall. The generated map supports residents and responsible authorities in identifying high-risk locations and planning suitable preventive actions. This component provides important location-based information for the overall Flood Guard 360 system.

3.4 Component 2 summary - Real-Time Drain Monitoring and Blockage Detection

The Real-Time Drain Monitoring and Blockage Detection component monitors drain water levels and rainfall conditions using IoT sensors connected to ESP32 microcontrollers. The collected readings are sent to the backend and displayed through the system dashboard. Three ultrasonic sensors installed at different points are used to compare water levels and identify unusual differences that may indicate a possible drain blockage. The system can generate alerts when configured conditions are detected, helping users identify potential problems earlier. This component supports flood prevention by improving visibility of drain conditions and providing information for timely action.

3.5 Component 3 summary - Post-Flood Disease Detection

The Post-Flood Disease Detection component aims to support the identification of possible health risks following a flood. Floodwater and contaminated surroundings can increase the risk of certain waterborne and vector-borne diseases. The component allows users to report relevant symptoms and uses the implemented analysis methods to identify possible health concerns. The results can help communicate health-risk information to users and support medical or public health teams in deciding where further attention may be required. The component is intended for preliminary risk screening and awareness, not for providing a confirmed medical diagnosis.

3.6 Component 4 summary - Post-Flood Rescue and Ration Distribution Management

The Post Flood Rescue and Ration Distribution Management component supports the coordination of rescue activities and essential supply distribution after a flood. It uses available information about affected locations and applies a priority classification approach to categorize locations into high, medium, and low priority levels. A dashboard and visual representations, such as heatmaps, can help responsible authorities understand where assistance may be needed most urgently. By combining relevant information from the other components, this component aims to improve rescue planning, resource allocation, and coordination during post flood recovery. Final decisions should also consider verified field information and the judgement of responsible authorities.

3.7 End to end user and data flow

Flood Guard 360: 24/7, Every Hour follows an end-to-end data flow in which information is collected, processed, analyzed, and presented to support flood risk management and post flood response. The four components work together by providing information about flood prone locations, drain conditions, possible health risks, and rescue and ration distribution priorities.

The process begins with data collection from different sources. The Flood Risk Map Creation component uses geographical information, rainfall data, and historical flood records to identify and display areas that may be at risk of flooding. At the same time, the Real Time Drain Monitoring and Blockage Detection component collects water level and rainfall readings through IoT sensors. The readings are transmitted to the backend and processed to identify unusual conditions, such as water level changes that may indicate a possible drain blockage. Relevant information is then presented through the dashboard, and alerts can be generated when configured conditions are met.

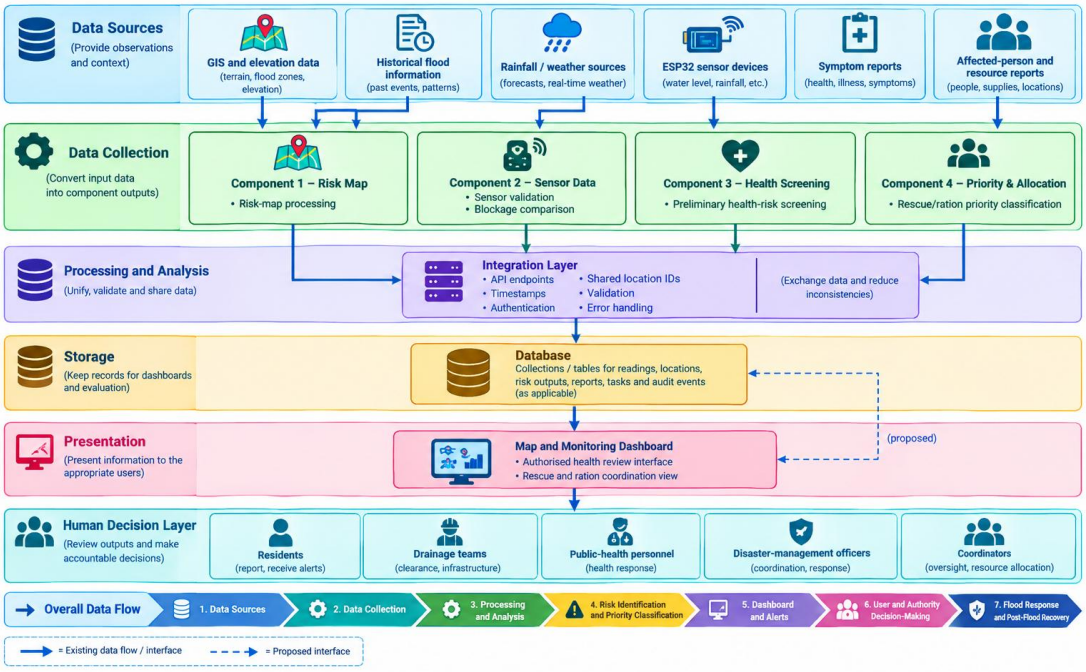
When a flood occurs, the Post Flood Disease Detection component supports the collection and analysis of reported symptoms to identify possible health risks. The Post

Flood Rescue and Ration Distribution Management component uses available information about affected locations to classify rescue priorities and support the planning of essential supply distribution.

The information produced by these components supports users and responsible authorities in understanding the situation and planning suitable actions. For example, flood risk maps and drain monitoring information can help identify locations that require attention, while health risk information and rescue priority classifications can support post flood response planning.

Overall, the end-to-end flow connects data collection, processing, risk identification, dashboard presentation, and decision making. The level of integration depends on the data exchanged between the components and the successful operation of the connected services. The system is intended to support decisions rather than replace field inspections, medical professionals, or emergency authorities.

Figure 2 : Data Flow



4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

The integration strategy of Flood Guard 360: 24/7, Every Hour focuses on bringing the four individual components together to support a complete flood management and post-flood response system. Each component has a specific responsibility, but the information produced by each component contributes to the overall purpose of identifying flood risks, monitoring changing conditions, and supporting timely decisions.

The system follows a modular integration approach. Each component is developed separately according to its own requirements and technologies. The Flood Risk Map Creation component provides geographical flood risk information, while the Real Time Drain Monitoring and Blockage Detection component collects sensor readings and identifies unusual drain conditions. The Post Flood Disease Detection component supports the identification of possible health risks, and the Post Flood Rescue and Ration Distribution Management component helps classify affected locations according to their assistance priorities.

The components exchange or use relevant information through their respective data processing and presentation mechanisms. Backend services process incoming data where implemented, and dashboards present relevant information to users. This modular approach allows each component to be developed, tested, and improved independently while supporting future integration with the overall system.

Data consistency and communication are important parts of the integration strategy. Information exchanged between components should use suitable data formats, clearly defined fields, and consistent location references where applicable. The system should also handle unavailable data, communication failures, and invalid sensor readings to reduce the risk of displaying misleading information.

Testing is carried out at both component and integration levels. Component level testing checks whether individual functions operate as expected, while integration testing checks whether the relevant data can be transferred, processed, and displayed

correctly across connected components. The integration process also considers the reliability of alerts and the usefulness of the information presented to users.

Overall, the integration strategy aims to create a modular and maintainable system in which the four components contribute to flood risk monitoring and post flood management. The level of integration achieved should be evaluated based on the implemented connections and test results, rather than assuming that every component is fully connected or operated in real time.

4.2 Interfaces data contracts and shared models

Flood Guard 360: 24/7, Every Hour consists of four components that handle different types of flood-related information. Interfaces and data contracts define how information is collected, transferred, validated, and used by these components. Shared models help maintain a consistent structure for important information, particularly location details, timestamps, risk levels, and status information.

Interfaces

The main interfaces of the system are described below.

- Flood Risk Map Creation: Uses geographical information, rainfall data, and historical flood records to identify and display potentially flood-prone locations.
- Real-Time Drain Monitoring and Blockage Detection: Uses ESP32-based IoT devices to collect water-level and rainfall readings. The readings are transmitted to backend API endpoints using HTTP POST requests.
- Post-Flood Disease Detection: Receives symptom information from users and processes it to identify possible health risks. Relevant results can be presented for review by authorized users.
- Post-Flood Rescue and Ration Distribution Management: Uses information about affected locations and priority classifications to support rescue coordination and ration distribution planning.

- Dashboard and user interfaces: Present relevant maps, sensor readings, risk information, alerts, and priority classifications according to the implemented functionality and user access permissions.

Data Contracts

A data contract defines the expected fields, data types, units, and validation rules for information exchanged between system components. The following table describes the main data requirements.

Table 4 : Data Contracts

Data category	Example fields	Purpose
Location data	Location ID, latitude, longitude	Identifies the relevant geographical area
Drain sensor data	Sensor ID, timestamp, water level, rainfall reading	Records current drain and rainfall conditions
Blockage data	Monitoring point IDs, water-level differences, blockage status	Identifies unusual water-level patterns that may indicate a blockage
Flood-risk data	Location ID, risk category, data source	Represents the estimated flood risk for a location
Health-risk data	Report ID, reported symptoms, screening result	Supports preliminary identification of possible health concerns
Rescue and ration data	Location ID, priority level, assistance status	Supports rescue planning and resource distribution

4.3 Shared nonfunctional requirements

Table 5 : Nonfunctional requirements

Requirement	Design approach	Evidence to collect
Reliability	Validate sensor values; detect disconnections; show last successful update; avoid treating missing readings as zero	Sensor fault and Wi-Fi/API outage test logs
Performance	Use an agreed sampling interval and monitor API response time and dashboard refresh delay	Timestamped measurements over repeated trials
Security	Authentication for protected routes, role-based access, secure secret handling and HTTPS for remote deployment	Access control checks and configuration review
Privacy	Collect minimum personal data, restrict health reports, define retention and deletion rules	Privacy review and role based access tests
Usability	Use clear labels, consistent status colors with text, understandable priority categories and accessible layouts	Task based usability feedback
Maintainability	Separate component services, document APIs, use version control and record configuration	Repository history, setup instructions and interface tests
Scalability	Use stable IDs, indexed queries and bounded sensor payloads; assess load before deployment	Load tests and database query measurements
Safety and accountability	Present alerts as decision support, preserve human review and log critical changes	Scenario walkthroughs and audit log checks

4.4 Integrated workflow and exception handling

Integrated Workflow

The workflow begins with the collection of information from different sources. The Flood Risk Map Creation component uses geographical information, rainfall data, and historical flood records to identify potentially flood prone areas. The Real Time Drain Monitoring and Blockage Detection component collects water level and rainfall readings using IoT sensors and sends the readings to the backend for processing.

The collected information is processed to identify relevant conditions. For example, the drain monitoring component compares water levels at different points to detect unusual differences that may indicate a possible blockage. When configured conditions are met, the system can generate alerts to inform users of potential problems. Flood risk information and drain monitoring results can help users identify locations that may require further attention.

After a flood, the Post Flood Disease Detection component processes reported symptoms to identify possible health risks. The Post Flood Rescue and Ration Distribution Management component uses available information about affected locations to classify assistance priorities and support the planning of rescue activities and ration distribution.

The results are presented through the relevant dashboards and interfaces, depending on the functionality implemented in each component. Users and responsible authorities can review the information and use it to support decisions. Any critical decision should also consider field observations and confirmation from the appropriate authorities.

Exception Handling

The system may encounter errors during data collection, communication, processing, or presentation. Appropriate exception-handling methods are needed to reduce the effect of these problems.

- Sensor failure: If a sensor provides invalid or missing readings, the system should identify the data as unreliable where possible and avoid treating it as a valid measurement.
- Network failure: If an IoT device loses its internet connection, data transmission may be interrupted. The system should indicate communication problems where this status is available and resume transmission when connectivity returns.
- Invalid data: Missing fields, unexpected values, and incorrect data formats should be checked before the information is processed or stored.
- Backend or database failure: If a backend service or database is unavailable, the system should handle the failed request appropriately and avoid reporting that the data was successfully stored without confirmation.
- Alert delivery failure: If an alert cannot be delivered, the failure should be recorded where possible. Users should not assume that an alert has been received unless delivery is confirmed.
- Unavailable flood-risk information: If required geographical or historical data are missing, the system should avoid presenting an unsupported risk classification as a confirmed result.
- Incomplete health information: If a user provides insufficient symptom information, the system should indicate the limitation rather than producing an unreliable health risk result.
- Unavailable rescue information: If current information about an affected location is unavailable, its priority should be reviewed using verified field information instead of relying only on outdated classifications.

4.5 Team contribution and dependency matrix

Table 6 : Team Contribution

Member	Primary component	Shared responsibilities
SOMARATHNE B.D(IT22091734)	Flood Risk Map Creation	Agree location IDs; test map data exchange; document data sources
BULUMULLA D.M.D.B (IT22278258)	Drain Management	Define sensor payloads; test backend integration; document readings and blockage validation
JAYANATH H.P.K.N(IT22170002)	Post-Flood Disease Detection	Agree privacy controls and location linkage; document screening limitations
RASHMI P.M.U(IT22130198)	Rescue and Ration Distribution Management	Agree priority inputs; test handover of affected location information
All team members	Integrated system	Interface agreement, integration testing, report review, source verification and final demonstration

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The entire system is made up of embedded sensors, services and database, GIS, internet (or web), and a web browser interface. The prototype drain monitoring system makes use of the ESP32-family microcontroller, ultrasonic sensors and a rainfall sensor. The data collected from the sensors is then passed onto the backend API via Wi-Fi to be processed. The output of the system can then be visualized and interacted with on a web dashboard, alongside a shared database which stores the output of each component. GIS tool and data sources are used to support the map feature, and the health monitoring and response components will either use a rule-based response or machine-learning model to guide actions. The final report documents the specific technologies and software versions, however only those technologies used by each team member in the final system are included.

Table 7 : Common Technology

System area	Known or intended technology	Implementation evidence to confirm
Drain hardware	ESP32-WROOM-32U and ESP32-S3; ultrasonic sensors; rainfall sensor	Board models, wiring diagrams, calibration notes and firmware version
Drain data transfer	Wi-Fi and HTTP POST to backend API routes	Successful response codes and backend/database records
Backend	API endpoints for sensor and blockage readings	Framework/version, route definitions, validation and error logs
Data storage	Database for time-stamped readings and component records	Schema, indexes, backup configuration and sample records

Risk mapping	GIS processing and geographic/environmental datasets	Software version, dataset licences, coordinate system and reproducible method
Health screening	Symptom-reporting workflow and documented rules or ML model	Model/rule specification, evaluation results, consent and access controls
Rescue and ration	Priority classification, potentially using Random Forest	Dataset, label definition, evaluation metrics and review process
Dashboard	Web-based component views and role-specific screens	Screenshots, user journeys, access controls and usability feedback

5.2 Development and integration process

The project was carried out through several stages, including requirements analysis, a literature review, component design, implementation, interface definition, system integration, testing, and refinement. Version control and issue-tracking tools were used to manage code changes and monitor development tasks. During integration, the team first tested small, individual data exchanges, such as sending a valid sensor payload, before moving on to testing multiple locations and complete system workflows.

The drain monitoring component involved calibrating the sensors, developing firmware to collect readings, establishing Wi-Fi communication, validating API payloads, storing the collected data, and displaying the results on the dashboard. The mapping component involved preparing source data, conducting spatial analysis, classifying risk levels, and presenting the results on a map. For the health monitoring and rescue components, the team documented the inputs, rules or models used, interpretation of outputs, and relevant safety checks. Evidence of the development process, such as repository history, task boards, and meeting notes, was included in the final report where available.

5.3 Deployment and operational configuration

The prototype was configured to improve security, reliability, and ease of maintenance. Sensitive information, such as API keys and database credentials, should be stored outside the source code wherever possible. For remote deployment, encrypted communication and authenticated API endpoints should be used to protect data. During local testing, the backend can run on a local machine; however, devices outside the local network cannot access it directly without the necessary network configuration.

The team documented the steps required to start the services, configure environment variables and the database, and monitor application logs. Operational documentation also covers device power requirements, sensor mounting positions, calibration procedures, sampling intervals, network requirements, backend ports, API endpoint paths, recovery procedures, and safe shutdown methods.

The drain monitoring component has used a separate port for blockage readings during development. This configuration must be verified against the final deployment environment and documented accurately to ensure that the system operates as expected.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

System testing focuses on verifying whether the different components communicate correctly and work together as expected. The team also evaluates whether the overall workflow is easy for the intended users to understand and whether errors or system failures are clearly displayed. Testing is carried out to identify communication issues, integration errors, and usability problems that may affect the system's performance.

The system is considered ready when the implemented components work together as intended, the main functions meet the defined requirements, and important failure conditions can be identified and handled appropriately. However, any features that have not yet been fully tested are reported as pending verification rather than being considered successful. Further testing and refinement will be carried out where necessary to improve the system's reliability and usability.

6.2 Integration and end to end testing

System testing focuses on verifying whether the different components communicate correctly and work together as expected. The team also evaluates whether the overall workflow is easy for the intended users to understand and whether errors or system failures are clearly displayed. Testing is carried out to identify communication issues, integration errors, and usability problems that may affect the system's performance.

The system is considered ready when the implemented components work together as intended, the main functions meet the defined requirements, and important failure conditions can be identified and handled appropriately. However, any features that have not yet been fully tested are reported as pending verification rather than being considered successful. Further testing and refinement will be carried out where necessary to improve the system's reliability and usability.

Table 8 : Integration test case

Test ID	Scenario	Expected behavior	Actual result / evidence	Status
IT-01	Verify water level and drain data is transmitted, stored, and visualized.	Readings are received, stored, and displayed as updated dashboard values and graphs.	Readings are received, stored, and displayed as updated dashboard values and graphs.	Pass
IT-02	Verify multiple data sources are processed together to generate flood threat zones.	Flood scenario with affected and unaffected areas	Flood map identifies affected areas and threat levels correctly.	Pass
IT-03	Verify processed flood risk results are correctly transferred and visualized.	Dashboard displays corresponding flood threat information and map output.	Dashboard displays corresponding flood threat information and map output correctly.	Pass
IT-04	Normal flow across three sensor points	System does not persistently report a blockage unless the calibrated rule is met	System does not persistently report a blockage unless the calibrated rule is met as expected.	Pass
IT-05	Controlled blockage introduced	Possible blockage is flagged according to the	Threshold value is correctly read along with the	Pass

		documented threshold and persistence rule	persistence rule to calculate the blockages correctly.	
IT-06	Risk map location linked to another component	Location identifiers match and map context is displayed correctly	Location identifiers match to the longitudes and latitude but doesnt exactly map to a actual city or location point name.	Fail
IT-07	Health report viewed by unauthorised role	Access is denied and sensitive information is not exposed	Access is denied	Pass
IT-08	Rescue priority reviewed and corrected	Authorised reviewer can understand, verify and correct the priority with an audit record	Working as expected	Pass
IT-09	One component unavailable	Other safe functions continue; missing component data are clearly marked	Working as expected	Pass
IT-10	End-to-end scenario from flood risk to response request	All implemented interfaces preserve location,	All implemented interfaces preserve location,	Pass

		timestamps and status correctly	timestamps and status correctly	
IT-11	Verify health inputs are processed together and produce a disease risk output.	System returns an appropriate disease risk, alert, or recommendation.	System returns an appropriate disease risk, alert, or recommendation.	Pass
IT-12	Verify end-to end operation of all major modules.	Relevant information is processed and available through the centralized platform without breaking module interactions.	All functions working correctly.	Pass

6.3 Overall results

Table 9 : Results

Area	Result to report	Evidence source
Flood-risk map	Study area Colombo District, Western Province, Sri Lanka Flood dataset sri_lanka_flood_dataset_district.csv Rainfall dataset	The study focuses on Colombo District, Sri Lanka, where the GIS-based flood-risk prediction component of Flood Guard 360 uses flood-related

	sri_lanka_rainfall_forecast_district.csv • Geographic scope- Colombo District only • Validation approach- 80% training / 20% testing, according to the training script • Models evaluated- Random Forest and Gradient Boosting • Risk categories- Low, Medium, High, Very High • Input features- Predicted rainfall, latitude, longitude, water level, flow rate, elevation and historical risk • Performance metrics- 80% training / 20% testing, passed as test_size=0.2	records, rainfall forecast information and geographical and hydrological features to classify flood risk into four categories: Low, Medium, High and Very High. Random Forest and Gradient Boosting models are evaluated using an 80:20 training-test split. Model performance is assessed using accuracy, precision, recall and F1-score. The evaluation results and dataset statistics are documented using the model training report.
Drain monitoring	Sampling interval, valid payload rate, sensor error and API/database success rate	Component 2 test log, firmware and backend records

Blockage detection	Number of normal and blockage trials, detection rate, false positives and limitations	Controlled test log and raw sensor readings
Disease screening	Dataset size and source, evaluation metrics, review protocol and limitations	Component 3 thesis and authorised evaluation records
Rescue/ration prioritisation	Study area: Colombo District, Western Province, Sri Lanka Rescue and relief dataset: camp_relief_priority_dataset.csv • **Geographic scope:** Colombo District only • **Dataset size:** 3,567 raw records with 36 columns. • **Validation approach:** 80% training / 20% testing • **Model evaluated:** Multi-Output Random Forest • **Priority categories:** Low, Medium, High • **Prediction targets:** Overall camp priority, food priority, water priority, medicine priority and sanitary-supply priority • **Input features:** Population; counts of children, elderly, infants, pregnant women, disabled people and chronic	Component 4 thesis and test records

	patients; vulnerable ratio; available food, water, medicine and sanitary stock; camp capacity; occupancy ratio; distance from distribution centre; hours since last distribution; road access status; vehicle capacity; latitude and longitude	
Integration	Number of end-to-end tests passed, failures, recovery behaviour and unresolved defects	Shared test matrix and issue log

6.4 Stakeholder and user validation

Stakeholder and user validation was conducted to evaluate the usability, understandability, and perceived usefulness of the Flood Guard 360 prototype. The evaluation focused on flood monitoring and dashboard visualization, flood prediction and threat mapping, post-flood rescue and ration distribution, disease management, and overall system usability. Feedback was collected from four participants representing different intended user perspectives: a doctor, a homeowner, a rescue team member, and a member of the disaster planning department.

6.4.1 Feedback collection method

The participants were presented with the Flood Guard 360 system, and their feedback was recorded using a Google Form on 29 August 2026. Five questions were used to assess overall system usability, the clarity and usefulness of flood monitoring information, the interpretability of the flood threat map, the usefulness of rescue and ration management information for decision-making, and the understandability of

disease management outputs. Responses were summarized using the ratings and qualitative feedback recorded in the user feedback report.

6.4.2 Evaluation results

The feedback indicated generally positive perceptions of the system's usability and information presentation. Overall ease of understanding and navigation received a rating of 5/5. The flood threat map also received 5/5, with 75% of participants rating it very easy to interpret and 25% rating it easy. Flood monitoring information and disease management outputs each received a rating of 4/5. For both features, 50% of participants rated the information very easy to understand and 50% rated it easy.

Rescue and ration management received a rating of 3/5, indicating a comparatively lower perceived level of usefulness among the evaluated features. Although the qualitative feedback did not identify a specific improvement for this component, the result indicates that its usefulness should be examined further in subsequent evaluations.

6.4.3 Improvements identified through feedback

The evaluation identified two principal areas for further development. First, feedback on disease management recommended improving disease detection accuracy and introducing disease severity assessment to help users better understand the potential threat associated with an identified illness. The feedback report records that a quality assurance (QA) component was added to help verify the accuracy of disease management outputs.

Second, a participant recommended adding house-address-based searching to the flood threat map to make location-specific information easier to access. According to the feedback report, implementation of this functionality was in progress at the time of reporting. These findings demonstrate how stakeholder feedback can inform the prioritisation of interface and system improvements.

6.4.4 Limitations of the evaluation

Although the feedback provides useful initial evidence of perceived usability and clarity, the evaluation involved only four participants. Consequently, the findings cannot be generalized to all intended users or emergency-response settings. The ratings represent participant perceptions and do not independently establish the predictive accuracy of the flood-risk model, the clinical validity of disease screening, or the operational effectiveness of rescue and ration allocation.

The available feedback report also does not fully document the participant recruitment procedure, detailed task-completion observations, consent process, or a formal qualitative analysis protocol. These details should be added if supporting records are available. Future evaluation should involve a broader and more representative group of stakeholders, structured task-based testing, documented consent procedures, and systematic analysis of usability, decision support, and uncertainty communication. Further testing should also assess whether users understand the limitations of predictions and recognize when independent verification or human review is required.

6.5 Achievement of common objectives

Table 10 : Achievements

Objective	Evidence required	Achievement status
Geographic flood-risk representation	GIS flood-risk map for Colombo District, flood and rainfall datasets, training_report.json, model validation metrics and map screenshots.	Partly achieved — The district-focused prediction workflow and mapping component are documented. Final achievement depends on confirming the generated map and actual evaluation results.
Real-time drain and rainfall monitoring	Sensor firmware, sampling configuration, API endpoints, stored sensor readings, rainfall data and communication logs.	Partly achieved — Monitoring functionality and related project components are described. Actual real-time operation and reliability need to be supported by sensor and backend test records.
Multi-point possible blockage detection	Controlled tests using normal and blocked drain conditions, raw sensor readings, detection rate, false-positive analysis and test screenshots.	Partly achieved — Blockage detection is part of the system scope, but controlled test results and false-positive measurements have not

		been established by the available evidence.
Preliminary post-flood health-risk workflow	Disease-screening workflow, dataset and label definitions, privacy review, model evaluation, QA evidence and limitations.	Partly achieved — The disease management component is documented, and the user feedback report records that a QA component was added. Clinical validity and screening performance require separate supporting evidence.
Rescue and ration prioritisation	Priority logic or model, dataset and labels, class-wise evaluation, prioritisation examples and documented human-review process.	Partly achieved — The component is included in Flood Guard 360, and stakeholder feedback was collected. Measured prioritisation performance and operational validation remain to be demonstrated.
Shared interfaces and end-to-end flow	API contracts, integration test matrix, request/response logs, database records, screenshots, failure-recovery tests and unresolved defect log.	Partly achieved — The integrated system architecture and user-facing functionality are documented. A complete set of verified end-to-end

		test results is still needed to establish full achievement.

6.6 Integrated limitations and threats

The reliability of Flood Guard 360 is influenced by data quality, sensor behavior, processing performance, network connectivity, model limitations, and the degree of integration between its components. As the system is a research prototype, the current implementation and testing do not establish that it will operate reliably under all geographical, environmental, and disaster conditions. The identified limitations and potential effects must therefore be considered when interpreting the system's results.

Flood-risk mapping. The flood-risk mapping component has a processing-time limitation, as map result generation may take a significant amount of time. This can delay the availability of predicted flood-risk information during time-sensitive situations. Furthermore, the model's outputs depend on the quality, completeness, and geographical representativeness of the historical flood and rainfall data. The current study focuses on Colombo District; consequently, the findings should not automatically be generalized to other districts with different geographical characteristics, drainage infrastructure, or rainfall patterns. Historical geographical data may also fail to represent recent changes to drainage systems and smaller drainage structures. Future work should optimize map generation, measure processing time under different workloads, and validate predictions using independent data and additional geographical areas.

IoT drain and rainfall monitoring. The prototype has identified occasional sensor data delays and fluctuations, which may cause dashboard values to lag behind current environmental conditions. Sensor calibration, positioning, communication reliability, and data validation can further influence the accuracy of displayed readings. For blockage detection, water-level differences may be affected by natural channel gradients, variations in drain geometry, incorrect sensor positioning, and changes in

water flow. Consequently, a detected abnormality may not always indicate a physical blockage. Repeated controlled trials using normal and blocked conditions, reference measurements, and different sensor positions are required to evaluate detection performance and false-positive behavior. Improvements to data transmission, validation, and signal smoothing should also be assessed.

Route planning and rescue coordination. Route planning previously produced incorrect routes, and the implemented correction requires further integration and regression testing using valid location data. Incorrect route information could affect rescue planning and delay the identification of suitable routes during a flood. Route reliability may also be affected by missing or inaccurate location information and changes to road accessibility during emergencies. Future testing should verify route outputs against known locations and realistic restrictions, including potentially inaccessible roads, and should confirm that corrected functionality remains reliable across different scenarios.

Post-flood health-risk management. The disease management component may not always distinguish diseases with similar symptoms or attributes correctly. Incomplete reports, uncertain labels, limited datasets, and differences between the data used during development and actual post-flood conditions may contribute to inaccurate or unclear health-risk outputs. Such outputs could lead to unnecessary alerts or failure to identify cases requiring attention. The prototype QA and verification functionality provide a development measure, but it does not independently establish clinical accuracy. Further evaluation should use appropriately governed data, clearly defined labels, additional test cases, and qualified domain review. Disease severity assessment should be developed and validated where appropriate. The component must remain a preliminary screening aid rather than a substitute for professional medical assessment.

Rescue and ration prioritisation. Automated prioritisation may be affected by incomplete incident reports, inaccurate location or victim information, class imbalance, and biases in the training data or prioritization rules. Performance may also deteriorate when the system encounters circumstances that differ from those represented during development. An incorrect priority assignment could influence

how assistance is allocated. Further work should evaluate class-wise performance, examine errors involving urgent cases, assess fairness where suitable data and methods are available, and document the reasons for prioritization decisions. Final rescue and ration allocation decisions should remain subject to appropriate human review.

Integration and evaluation coverage. Testing evidence currently reflects the prototype and the scenarios evaluated rather than the full range of possible disaster conditions. The availability of working individual components does not, by itself, demonstrate that the complete system will function reliably from sensor input through backend processing to dashboard presentation and response workflows. Failures may occur at component boundaries, particularly during network interruptions, delayed data transmission, invalid inputs, or service recovery. End-to-end testing should therefore cover normal operation, invalid inputs, communication failures, recovery behaviour, and unresolved defects. Independent ground-truth data and broader testing conditions are also needed to strengthen confidence in the reported results.

Stakeholder and operational limitations. User feedback was collected from four participants representing healthcare, household, rescue, and disaster-planning perspectives. The feedback provides preliminary evidence about perceived usability and information clarity, but the small participant group is not sufficient to establish usability across all intended users or emergency-response settings. Further validation should include a larger and more diverse group of stakeholders, structured task-based testing, and documented evaluation procedures. Users must also be made aware that predicted risk categories, sensor indicators, health-screening outputs, and priority labels may contain errors or uncertainty. Over-reliance on automated outputs could result in inappropriate decisions if those outputs are treated as definitive.

Mitigation and future work. The project's limitation register identifies map-processing performance and IoT data consistency as areas requiring improvement, while disease differentiation remains open for further development and route accuracy requires retesting. These limitations define the current boundaries of the implementation rather than automatically invalidating the prototype's results. Future work should prioritise the outstanding mitigation actions, expand controlled and end-

to-end testing, use independent validation data where available, communicate uncertainty clearly, protect sensitive information, and maintain appropriate human oversight. Gradual evaluation under realistic conditions is necessary before considering wider operational deployment.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The reliability of the system depends on several factors, including data quality, sensor calibration, network connectivity, and the level of integration between its components. Since the system is developed as a prototype, testing under limited conditions is not sufficient to confirm that it will perform reliably across different drain designs, rainfall intensities, and geographical areas. For example, blockage detection based on water-level differences may be affected by natural channel gradients or incorrect sensor positioning. Similarly, historical geographical data may not reflect recent changes to drainage systems or smaller drainage structures.

The health monitoring component may also be affected by incomplete or biased reports, while automated screening may generate false alarms or fail to identify certain cases. Likewise, a rescue-priority model may inherit biases from historical data or produce unreliable results when applied to situations that differ from its training data.

Additional limitations include small test samples, the absence of independent ground-truth data, differences between simulated and actual flood conditions, limited diversity among participants, and changes in model performance over time. There is also a risk that users may place too much trust in automated outputs without considering their limitations. These challenges can be addressed through repeated controlled testing, independent validation, clear communication of uncertainty, appropriate human supervision, privacy protection, and gradual deployment under real-world conditions. Further testing and evaluation are necessary before the system can be considered reliable for wider use.

Potential user	Possible value	Evidence needed
Residents	Clearer awareness of local risk and status	Comprehension and usefulness feedback
Drainage teams	Earlier visibility of abnormal local readings and possible obstructions	Comparison with inspection findings and response time
Local authorities	A geographic overview to support planning	Map validation and operational workflow evaluation
Public-health teams	Structured reports for review and follow-up	Expert review, privacy assessment and safe referral workflow
Rescue and relief coordinators	A consistent way to review requests and track allocation	Scenario testing, auditability and responder feedback

7.2 Market alternatives and differentiation

Flood management can involve several existing approaches, including manual drain inspections, sensor-based monitoring systems, standalone flood-risk maps, public warning services, health-reporting workflows, and manual coordination of rescue and relief activities. These approaches address different operational needs and may already be effective within their respective areas of application.

Flood Guard 360 differentiates itself through its proposed integration of multiple flood management functions within a common framework. The system connects pre-flood risk representation and monitoring with post-flood health-risk screening and rescue and ration prioritization. This integrated approach is intended to improve information accessibility and coordination across the flood management lifecycle.

The main areas of differentiation are summarised below.

Table 11 : Main areas of differences

Evaluation dimension	Existing or conventional approaches	Flood Guard 360 approach
Flood-risk information	May be provided through separate maps, reports, or warning services	GIS-based flood-risk representation within the wider platform
Drain monitoring	May rely on periodic physical inspection or dedicated monitoring systems	IoT-based monitoring of drain-related readings and possible abnormal conditions
Post-flood health management	May depend on separate reporting and follow-up processes	A preliminary health-risk screening workflow
Rescue and ration coordination	May use manual records and separate prioritisation processes	Structured incident information and prioritisation support
Information integration	Depends on the systems and communication procedures used by each organisation	Designed to connect information across the four project components

7.3 Business model and feasibility

Flood Guard 360 has potential as a technology solution for organisations involved in flood preparedness, drainage maintenance, disaster response, public-health coordination, and relief distribution. Potential adopters include local authorities, disaster management organisations, drainage service providers, and relevant public-sector agencies. However, commercial demand, willingness to pay, and the financial viability of the proposed solution have not yet been established.

A possible business model is a business-to-government or institutional service in which an organisation adopts the platform to support its flood management activities. Depending on the deployment requirements, the service could be provided as a

centrally hosted platform, a locally deployed system, or a combination of software and field-monitoring equipment.

Potential delivery and revenue approaches include:

- **Institutional licensing:** An annual licence or service agreement for access to the platform and its supported modules.
- **Implementation and integration services:** Configuration, data integration, dashboard customisation, and connection to existing organisational systems.
- **IoT installation and maintenance:** Provisioning, installation, calibration, repair, and replacement of sensors and associated equipment.
- **Training and technical support:** Training for operators and administrators, troubleshooting, system updates, and ongoing maintenance.
- **Optional managed services:** Hosting, monitoring, backups, and technical administration where these services are required.

7.4 Adoption intellectual property and compliance

The adoption of Flood Guard 360 depends on more than the availability of its software. Practical implementation may be affected by sensor damage, vandalism, limited network connectivity, shortages of trained operators, maintenance requirements, and uncertainty regarding responsibility for equipment and system support. Successful adoption would therefore require clear operational procedures, defined responsibilities, appropriate training, and agreements concerning data sharing and incident response.

Intellectual property management is another important consideration. The project should maintain an inventory of its source code, interface designs, model artefacts, documentation, datasets, libraries, and third-party services. The ownership and permitted use of these materials should be documented, and the licences of open-source software and external datasets should be checked before redistribution or commercial use. Any external mapping, hosting, or data services should also be reviewed for applicable usage restrictions and costs. No claim of patent protection,

registered intellectual property, or exclusive ownership should be made without supporting documentation.

Responsible data management is particularly important because the platform may process incident locations, household information, and health related reports. Access should be restricted according to user roles, sensitive records should be protected against unauthorized disclosure, and data collection should be limited to information necessary for the intended purpose. Appropriate consent or other lawful grounds, retention periods, deletion procedures, security controls, and data sharing arrangements should be established before operational deployment.

7.5 Sustainable Development Goal contribution

The adoption of Flood Guard 360 depends on more than the availability of its software. Practical implementation may be affected by sensor damage, vandalism, limited network connectivity, shortages of trained operators, maintenance requirements, and uncertainty regarding responsibility for equipment and system support. Successful adoption would therefore require clear operational procedures, defined responsibilities, appropriate training, and agreements concerning data sharing and incident response.

Intellectual property management is another important consideration. The project should maintain an inventory of its source code, interface designs, model artefacts, documentation, datasets, libraries, and third-party services. The ownership and permitted use of these materials should be documented, and the licenses of open-source software and external datasets should be checked before redistribution or commercial use. Any external mapping, hosting, or data services should also be reviewed for applicable usage restrictions and costs. No claim of patent protection, registered intellectual property, or exclusive ownership should be made without supporting documentation.

Responsible data management is particularly important because the platform may process incident locations, household information, and health related reports. Access should be restricted according to user roles, sensitive records should be protected

against unauthorized disclosure, and data collection should be limited to information necessary for the intended purpose. Appropriate consent or other lawful grounds, retention periods, deletion procedures, security controls, and data-sharing arrangements should be established before operational deployment.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

Flood Guard 360: 24/7, Every Hour aims to improve flood management by bringing together different activities involved before, during, and after a flood. The system consists of four main components: geographic flood-risk mapping, drain monitoring using water-level and rainfall sensors, preliminary health risk reporting after floods, and prioritization of rescue operations and ration distribution. The proposed architecture provides a common framework for sharing location and event-related information while allowing each component to perform its specific functions.

The drain monitoring prototype demonstrates the practical implementation of ESP32-based sensors, API communication, and the comparison of readings collected from multiple points to identify possible drain blockages. However, the effectiveness of this approach depends on proper sensor calibration and further controlled testing. Overall, Flood Guard 360 remains a research prototype until the integration of all components. Implementation of safety measures, and end-to-end system functionality have been thoroughly verified. Further testing and refinement will be necessary to assess its reliability and suitability for real-world flood management.

8.2 Principal integrated contribution

The main contribution of Flood Guard 360 is the development of a common framework that brings together information relevant to different stages of flood management. The framework supports communication between the system components through consistent location identifiers, time-stamped data, clearly defined statuses and measurement units, and appropriate data validation. It also establishes clear responsibilities for each component and emphasizes the importance of human review when interpreting system outputs.

8.3 Future directions

1. Completion and validation of system integration. The interfaces between all four components should be reviewed and verified using consistent data formats, location identifiers, timestamps, and status definitions. End-to-end tests should examine the complete flow of information from data collection and processing to dashboard presentation and response workflows. Testing should include invalid inputs, communication interruptions, service failures, recovery behavior, and the recording of unresolved defects.

2. Improvement of drain monitoring and blockage detection. The IoT monitoring component requires longer-duration testing and controlled experiments under normal-flow and simulated blockage conditions. Evaluation should measure detection rates, false-positive rates, sensor-data delays, and consistency across multiple monitoring points. Sensors should be calibrated for their intended installation positions, while weather-resistant enclosures, suitable power management, and reliable communication mechanisms should be assessed before outdoor deployment. Additional testing should determine how drain geometry, natural water-level differences, and sensor positioning affect detection results.

3. Further evaluation of flood-risk mapping. The GIS-based prediction component should be evaluated using independent historical records and, where available, verified flood observations. Validation should examine model performance, the reliability of the underlying datasets, geographical coverage, and the suitability of the selected risk categories. Further investigation should also assess prediction performance over different time periods and locations before extending the approach beyond Colombo District. Map-generation processing time should be measured and optimized where necessary.

4. Strengthening health-risk screening. The post-flood health-risk workflow should undergo further evaluation using appropriately governed datasets and review by qualified health professionals. Future development should investigate disease differentiation, severity assessment, and the handling of incomplete or uncertain reports. Any image-based features should remain outside operational use until their

accuracy, data-protection requirements, informed-consent arrangements, and applicable ethical approvals have been adequately addressed. The component should continue to be presented as a preliminary screening aid rather than a replacement for professional medical assessment.

5. Evaluation of rescue and ration prioritization. The prioritization component should be tested using representative scenarios and suitable evaluation datasets. Performance should be reported separately for relevant classes, with particular attention to errors affecting urgent requests. Fairness, explainability, and the handling of incomplete incident information should also be assessed. Appropriate human review should remain part of the workflow so that automated priority labels do not independently determine rescue or ration allocation.

6. Security, privacy, and stakeholder validation. Further work should include role-based access testing, security assessments, privacy reviews, and usability evaluations involving a broader range of intended users. Stakeholder evaluation should document participant roles, recruitment procedures, consent where applicable, task completion, feedback analysis, and changes made in response to the findings. The interface should communicate uncertainty and limitations clearly, particularly for flood-risk predictions, health-risk screening, and response priorities.

7. Supervised pilot deployment and operational sustainability. Once the necessary safety and reliability requirements have been addressed, a small, supervised pilot could be conducted in collaboration with a relevant authority or community organization. Before such a pilot, the project should establish responsibilities for maintenance, alert handling, data management, and technical support. Practical documentation should cover installation, calibration, troubleshooting, user training, estimated operating costs, data-retention procedures, and incident-response arrangements.

In conclusion, these future directions provide a pathway for progressing Flood Guard 360 from a research prototype towards a more thoroughly evaluated flood management platform. Its wider adoption should depend on demonstrated technical reliability,

appropriate safeguards, meaningful stakeholder feedback, and evidence that the integrated system can support decision-making under realistic operating conditions.

REFERENCES

[1] A. Rajasekar, M. Suman, J. S. Jai Aditya, S. Danush, and K. Shyam Sundar, "Drain block detection and controlling system," in *Proc. 2024 Int. Conf. Commun., Comput. Internet of Things (IC3IoT)*, 2024, doi: 10.1109/IC3IoT60841.2024.10550345.

[2] V. S. Gawali, M. Pande, M. Sayyad, and R. S. Bhadade, "IoT based real time drainage monitoring system," in *Proc. 2024 Int. Conf. Emerging Smart Comput. Informat. (ESCI)*, Pune, India, 2024, pp. 1–6, doi: 10.1109/ESCI59607.2024.10497198.

[3] T. Jayapriya, L. Bhuvaneswari, A. Asmitha, S. Durka, K. Suman, and R. Gokila, "IoT based smart drainage monitoring and alert system using mobile application integration," in *Proc. 2025 4th Int. Conf. Innovative Mechanisms Ind. Appl. (ICIMIA)*, 2025, pp. 227–232, doi: 10.1109/ICIMIA67127.2025.11200687.

[4] S. Chakraborty, S. Chatterjee, A. Sarkar, S. Kanjilal, G. Das, and P. Kundu, "Study of IoT enabled water logging detection and control in drainage system for smart city," in *Proc. 2024 Int. Conf. Big Data Analytics Bioinformat. (DABCon)*, 2024, doi: 10.1109/DABCON63472.2024.10919387.

[5] M. Z. S. Hadi, V. R. Putri, and M. Yuliana, "Design of water level and drainage monitoring system based on Internet of Things," in *Proc. 2023 IEEE Int. Conf. Commun., Netw. Satell. (COMNETSAT)*, Malang, Indonesia, 2023, pp. 570–576, doi: 10.1109/COMNETSAT59769.2023.10420559.

[6] A. Diriyana, U. Darusalam, and N. D. Natasha, "Water level monitoring and flood early warning using microcontroller with IoT based ultrasonic sensor," *Jurnal Teknik Informatika C.I.T*, vol. 11, no. 1, pp. 22–28, 2019.

[7] J. G. Natividad and J. M. Mendez, "Flood monitoring and early warning system using ultrasonic sensor," *IOP Conf. Ser.: Mater. Sci. Eng.*, vol. 325, Art. no. 012020, 2018, doi: 10.1088/1757-899X/325/1/012020.

[8] W. Indrasari, B. H. Iswanto, and M. Andayani, "Early warning system of flood disaster based on ultrasonic sensors and wireless technology," *IOP Conf. Ser.: Mater. Sci. Eng.*, vol. 335, Art. no. 012005, 2018, doi: 10.1088/1757-899X/335/1/012005.

[9] S. Dswilan, Harmadi, and Marzuki, "Flood monitoring system using ultrasonic sensor SN-SR04T and SIM 900A," *J. Phys.: Conf. Ser.*, vol. 1876, Art. no. 012003, 2021, doi: 10.1088/1742-6596/1876/1/012003.

[10] B. E. Pengel *et al.*, "Flood early warning system: sensors and internet," in *Floods: From Risk to Opportunity*, A. Chavoshian and K. Takeuchi, Eds., IAHS Publ. 357. IAHS Press, 2013, pp. 445–453.

[11] K. I. Dissanayake, K. D. J. Fernando, C. Weerarathne, D. Kavinda, and H. G. B. Chathuranga, "Design and experimental evaluation of an IoT-enabled semi-autonomous platform for sediment level detection and 3D mapping in urban canals," in *Proc. 2026 Moratuwa Eng. Res. Conf. (MERCon)*, 2026, doi: 10.1109/MERCon71835.2026.11691390.

[12] S. P. M. K. W. Ilukkumbure, V. Y. Samarasiri, M. F. Mohamed, V. Selvaratnam, and U. U. S. Rajapaksha, "Early warning for pre and post flood risk management by using IoT and machine learning," in *Proc. 2021 3rd Int. Conf. Adv. Comput. (ICAC)*, 2021, pp. 252–257, doi: 10.1109/ICAC54203.2021.9671141.

[13] Z. Farzan, "833,000 affected, death toll rises to 159 in Sri Lanka's worst floods in decades," *News 1st*, Nov. 30, 2025. [Online]. Available: https://newsfirst.lk/2025/11/30/833-000-affected-death-toll-rises-to-159-in-sri-lanka-s-worst-floods-in-decades [Accessed: Oct. 7, 2026].

[14] "Sri Lanka: Over 120,000 affected by severe flooding in west," *IRIN / The New Humanitarian*, May 2007. [Online]. Available: https://www.thenewhumanitarian.org/news/2007/05/07 [Accessed: Oct. 7, 2026].

[15] "Development of an integrated urban drainage cycle management model for city of Colombo," in *Proc. KDU Int. Res. Conf.*, 2014. [Online]. Available: https://ir.kdu.ac.lk/handle/345/1529 [Accessed: Oct. 7, 2026].

[16] A. Abeysinghe, "Flooding in Colombo aggravated by human activity," *AsiaNews*, Sep. 13, 2023. [Online]. Available: https://www.asianews.it/en/south-asia/sri-lanka/flooding-in-colombo-aggravated-by-human-activity [Accessed: Oct. 7, 2026].

Computer Vision and IoT-Based Sensors in Flood Monitoring and Mapping: A Systematic Review. (2019). Sensors, 22.

Demissie, Z. ,. (2024). Flood susceptibility mapping: Integrating machine learning and GIS for enhanced risk assessment. Applied Computing and Geosciences.

H., F. J. (2001). Greedy function approximation: A gradient boosting machine. The Annals of Statistics, 1189-1232.

L., B. (2001). Random forests. Machine Learning, 5-32.

Somarathne B.D, R. P. (2026). An Integrated AI-, IoT- and GIS-Based Platform for Flood-Risk Prediction, Real-Time River Monitoring, Post-Flood Disease Screening and Relief Distribution Management. 6.

APPENDICES

Appendix A Integrated interface specification

A.1 Purpose and scope

This appendix presents the interface specification for Flood Guard 360, an integrated flood-disaster management platform developed by research group R26-IT-031. The platform comprises four interconnected components: GIS-based flood-risk prediction and mapping, IoT-based drain and rainfall monitoring with possible blockage detection, post-flood disease-risk management, and rescue and ration distribution management.

The purpose of the specification is to document the technologies, information flows, interface requirements, and integration scenarios relevant to the complete system. It also identifies the verification activities required to assess whether the components exchange information correctly. The specification reflects the documented system design; it does not imply that every interface or end-to-end test has been independently verified

A.2 Integrated technology stack

Table 12 : **Integrated technology stack**

System layer	Technology or component	Purpose
Frontend	React.js, TypeScript, Tailwind CSS and Vite	Web application, dashboards and user interfaces
Geographic visualisation	Leaflet.js / React-Leaflet, OpenStreetMap and GeoJSON	Interactive flood-risk and location visualisation
Backend	Node.js and Express.js	API services and coordination of application requests
Real-time communication	WebSocket / Socket.IO	Time-sensitive status updates and alert communication
Machine learning	Python and scikit-learn	Prediction, classification and forecasting services

IoT hardware	ESP32-based monitoring devices, ultrasonic sensors and rainfall sensors	Collection of environmental and drain-monitoring measurements
Communication	Wi-Fi, HTTP and applicable messaging protocols	Transmission of sensor measurements and application data
Database	MongoDB	Storage of application records and structured component information
Security	JWT, bcrypt and role-based access control	Authentication, password protection and access management

The listed technologies describe the integrated architecture documented in the project materials. The exact implementation and configuration of each interface should be verified against the corresponding source code and deployment settings.

A.3 Component interface specification

Table A.1: Component interfaces and information exchange

Interface ID	Source	Destination	Information exchanged	Purpose
IF-01	Flood and rainfall datasets	Component 1: Flood-risk mapping	Historical flood records, rainfall information and geographical features	Prepare model inputs and generate geographical flood-risk predictions
IF-02	ESP32 monitoring devices	Component 2 backend services	Water-level readings, rainfall measurements, timestamps and monitoring-point information	Support drain monitoring and identification of possible abnormal conditions

IF-03	Component 1 prediction service	Geographic dashboard and integration layer	Location-associated flood-risk predictions and risk categories	Display predicted flood-risk information geographically
IF-04	Component 2 monitoring services	Dashboard and connected response workflows	Sensor readings, monitoring status and possible blockage alerts	Support monitoring and communicate abnormal drain conditions
IF-05	Health-reporting interface	Component 3: Disease-risk management	Symptom information, relevant risk factors and optional image inputs where enabled	Generate preliminary health-risk screening outputs
IF-06	Component 3 inference services	Health dashboard and authorised review workflow	Risk classifications and configured image-classification outputs	Support preliminary screening and healthcare review
IF-07	Public incident reporting and camp management interfaces	Component 4: Rescue and ration management	Rescue requests, incident locations, camp population, vulnerability information and available resource stocks	Organise incidents and assess rescue and relief requirements
IF-08	Component 4 prioritisation service	Rescue and ration management dashboard	Overall camp priority and resource-specific priority outputs	Support officer review and relief distribution planning
IF-09	Backend services and shared data layer	Integrated application workflows	Validated records, component outputs and relevant status information	Support information exchange and coordination between connected components

The interfaces represent the logical information flows described in the project documentation. Actual endpoint paths, HTTP methods, request and response schemas, and authentication requirements should be obtained from the implemented services before this appendix is treated as a complete API specification.

A.4 Common interface requirements

The following requirements support consistent information exchange between the components.

Table 13 : Common interface requirements

Requirement	Specification
Location identification	Use consistent location identifiers or documented geographic coordinates wherever applicable
Timestamps	Associate sensor readings, predictions and incident records with their relevant collection or processing times
Input validation	Validate required fields, data types, permitted values and measurement units before processing
Output interpretation	Define the meaning of flood-risk categories, sensor states, health-risk outputs and response priorities
Error handling	Identify invalid inputs, failed requests, unavailable services and unsuccessful processing
Data persistence	Verify that records intended for storage can be saved and retrieved correctly
Authentication and authorisation	Apply the implemented access controls to protected application functions and sensitive information
Privacy	Restrict access to personal, location-related and health-related records according to authorised roles
Traceability	Retain appropriate records of requests, outputs, failures and corrective actions where supported

A.5 Integrated operational scenarios

The following scenarios describe how the components are intended to exchange information within Flood Guard 360.

1. **Flood-risk generation:** Historical flood, rainfall and geographical information are prepared for the prediction service. The resulting risk classification is associated with a geographical location and presented on the map.
2. **Drain monitoring:** An IoT monitoring device collects water-level and rainfall measurements and transmits the readings to the backend. The monitoring service processes the received information and makes relevant readings or possible blockage alerts available to the dashboard.
3. **Possible blockage identification:** Water-level measurements from multiple monitoring points are compared. A persistent abnormal difference can trigger a possible blockage indication according to the configured detection logic. The indication requires confirmation and should not automatically be interpreted as a verified physical obstruction.
4. **Health-risk screening:** A user or authorised operator submits relevant health-related information. The disease management component generates a preliminary screening result for appropriate review. The output is not a confirmed clinical diagnosis.
5. **Rescue and ration prioritisation:** Incident and relief-camp information is processed by the rescue and ration management component. Overall camp priority and individual resource priorities are presented to authorised officers for review before operational decisions are made.
6. **Response coordination:** Relevant location, status and accessibility information can support rescue and relief planning. Completed rescue or distribution actions should be recorded where the implemented workflow supports this functionality.

These scenarios define the intended information flow and provide a basis for integration testing. They should not be interpreted as proof that every scenario has passed an end-to-end test.

A.6 Integrated verification test matrix

Table 14 : **Proposed integration verification tests**

Test ID	Test scenario	Expected result	Evidence to retain
INT-01	Submit valid rainfall and flood-related data	Valid data is accepted and made available to the flood-risk processing workflow	Input dataset, processing output and relevant logs
INT-02	Generate a geographical flood-risk classification	The prediction output is associated with the correct location and risk category	Model output and map screenshot
INT-03	Submit normal drain-monitoring readings	Readings are transmitted and processed without generating an unjustified critical blockage alert	Sensor readings, API response and stored records
INT-04	Submit abnormal readings from multiple monitoring points	The configured detection logic evaluates the readings and produces the appropriate possible blockage status	Raw readings, threshold configuration and alert record
INT-05	Submit a health-risk screening request	The service returns the configured screening output and handles invalid inputs appropriately	Request, response and relevant test log

INT-06	Submit relief-camp population and stock information	Overall and resource-specific priorities are generated according to the implemented model or logic	Input record, prediction output and dashboard evidence
INT-07	Test an unauthorised operation	The implemented access controls deny access where permission is required	Test account, request result and security log
INT-08	Simulate a communication or service failure	The failure is handled appropriately and recovery behaviour is recorded	Error log, recovery record and retest result
INT-09	Complete a connected disaster-response workflow	Information passes through the relevant connected components consistently	End-to-end execution log and screenshots

For each test, the actual execution date, test inputs, observed result, pass/fail status, evidence reference and unresolved defects should be recorded. Tests that have not been executed must be marked **Not tested** rather than reported as passed.

A.7 Integration limitations

The integrated platform remains subject to limitations involving sensor consistency, network connectivity, model performance, data completeness and communication between services. The implementation of an individual component does not independently establish that the complete system functions reliably under all operating conditions.

Further verification should concentrate on data consistency, location mapping, failure recovery, route-planning behaviour, health-screening limitations, and the correct

presentation of rescue and ration priorities. The test matrix should be updated as additional evidence becomes available. Any claim of successful end-to-end integration should be supported by the corresponding test records.

Appendix B Team contribution declaration

B.1 Purpose

This appendix records the individual and shared contributions of the four members of research group R26-IT-031 to the development of Flood Guard 360. The declaration identifies each member's principal component and summarizes the associated research, design, implementation and evaluation responsibilities.

Table 15 : Individual Contribution

Team member	Student ID	Principal component	Summary of contribution
Somarathne B. D.	IT22091734	Component 1: GIS-based flood-risk prediction and mapping	Research and development of the flood-risk mapping component, including geographical representation of predicted risk, preparation of relevant model inputs and integration of prediction outputs with the map interface.
Bulumulla D. M. D. B.	IT22278258	Component 2: Real-time drain and rainfall monitoring	Development of the IoT drain-monitoring component, including

			ESP32-based firmware, ultrasonic sensor readings, multi-point water-level comparison, possible blockage-detection logic and transmission of readings to backend services.
Jayanath H. P. K. N.	IT22170002	Component 3: Post-flood disease-risk management	Research and development of the symptom-based health-risk screening workflow and image-classification service, including model evaluation, inference-service integration, output validation and documentation of clinical limitations.
Rashmi P. M. U.	IT22130198	Component 4: Post-flood rescue and ration distribution management	Research and development of the rescue-request and relief-camp workflows, resource-specific prioritisation, officer-review processes and integration of relevant incident and accessibility information into response planning.

The allocation above reflects the principal components identified in the individual thesis documents. It does not imply that each component was developed without collaboration or that every planned feature has been fully validated.

B.3 Shared contributions

The following activities form part of the team's collective work where supported by the project records:

- Establishing the overall Flood Guard 360 objectives, requirements and integrated architecture.
- Coordinating information exchange and interface requirements between the four components.
- Integrating individual components with the shared backend, database and web application where implemented.
- Reviewing system behavior, troubleshooting technical issues and identifying unresolved defects.
- Preparing the integrated research documentation, project demonstrations and supporting evidence.
- Reviewing the system's limitations, privacy and security considerations, responsible use of automated outputs, and future development requirements.